Project Overview – Leadership Analytics Employee Attrition Prediction

The project is designed to analyze leadership-related employee data and predict **employee attrition risk** using machine learning techniques. The main steps of the project are summarized as follows:

1. Data Cleaning

- Removed missing values from the dataset to ensure reliable analysis.
- Checked the dataset for inconsistencies and ensured all variables were valid and usable.

2. Data Processing (Encoding)

- Converted categorical variables (**Department, JobRole**) into numerical values using **Label Encoding**.
- Ensured all features were in a format suitable for machine learning models.

3. Feature Engineering

- Created a new feature called **Leadership_Gap**.
- This feature represents the difference between **Performance Rating** and **Job Satisfaction**.
- Positive values may indicate employees who perform well but experience low satisfaction, which may signal potential leadership or management issues.

4. Machine Learning Models

Three machine learning models were implemented to analyze employee attrition:

- **Random Forest Classifier**
- **Logistic Regression**

- **Neural Network (MLPClassifier)**

Key features used in the models include:

- Age
- Department
- Job Role
- Job Satisfaction
- Monthly Income
- Training Times Last Year
- Years at Company
- Work Life Balance
- Performance Rating
- Leadership_Gap

5. Model Training

- The dataset was split into **training (80%)** and **testing (20%)** sets.
- Feature scaling was applied for models that require normalized data (**Neural Network and Logistic Regression**).
- **GridSearchCV** was used to identify the best hyperparameters for the Random Forest and Neural Network models.

6. Model Evaluation

The models were evaluated using several performance metrics:

- **Accuracy Score**
- **Mean Squared Error (MSE)**
- **Classification Report**
- **Confusion Matrix**

Additional visualizations were generated to interpret model performance, including:

- **Feature Importance Graph**
- **Confusion Matrix Heatmaps**

7. Results Interpretation

- The results showed that **Random Forest achieved the highest prediction accuracy**.
- Important factors affecting employee attrition included:
    - Job Satisfaction
    - Work Life Balance
    - Years at Company
    - Leadership_Gap
- Neural Networks were able to capture complex patterns in the data, while Logistic Regression provided a simpler and interpretable baseline model.

8. Report and Documentation

- All steps of the project were documented, including data preprocessing, model training, evaluation, and interpretation.
- Visualizations and explanations were provided to clearly present insights derived from the analysis.

Overall Conclusion

This project demonstrates how **Leadership Analytics combined with Machine Learning** can help organizations predict employee attrition, understand leadership-related factors affecting employee satisfaction, and support **data-driven HR decision-making** to improve employee retention and organizational performance.